Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Michael Miller	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University Dept. of Biomedical Engineering			
Address	3400 N. Charles Street Baltimore, MD 21218			

2. Technology Overview

Title of Proposed Technology	Clinically Deployable, Modality-Agnostic AI Platforms for Early Detection of Alzheimer's Disease			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic and Diagnostic Devices	Intelligent Interpretation System	Medical Information & Systems	Other Medical Information & Systems
Concept of Proposed Technology	Early and accurate diagnosis of neurodegenerative diseases such as Alzheimer's disease (AD) has become increasingly critical with the emergence of disease-modifying therapies. However, current diagnostic pathways rely heavily on costly, invasive, or infrastructure-intensive biomarkers, limiting scalability and equitable access in routine clinical care. This joint R&D program seeks to support the development of clinically deployable AI technologies that leverage routine neuroimaging data and complementary biomarkers to enable early disease detection, risk stratification, and prognosis prediction. Core principles of the proposed technology concept include: • Modality-agnostic AI architectures capable of robust performance across CT, 2D MRI, 3D T1, T2 MRI, or other commonly acquired imaging data; • Extraction of disease-relevant structural and functional signatures from standard-of-care imaging without dependence on specialized research protocols;			

(max 10 pages)

	• Integration of imaging-derived features with demographic, clinical, and minimally invasive biomarker data to enhance diagnostic confidence and enable precise risk stratification for disease progression. Proposals may address a range of methodological approaches, including but not limited to advanced segmentation, representation learning, domain adaptation, multimodal data fusion, and risk stratification frameworks. The RFP intentionally does not prescribe a specific algorithmic strategy, biomarker, or disease stage, allowing applicants to propose innovative and generalizable solutions.
Rationale for Joint R&D Project	Despite rapid advances in AI and biomarker science, translation of neuroimaging-based AI tools into routine clinical practice remains limited due to challenges in generalizability, regulatory validation, and real-world clinical integration. The rationale for a Korea–US joint R&D framework is as follows: • Korean startups/SMEs demonstrate strengths in agile AI development, data engineering, and productization but often lack access to diverse international clinical datasets and U.S.-based validation environments. • U.S. academic and clinical research institutions offer deep expertise in neurodegenerative disease biology, heterogeneous patient cohorts, multi-center validation, and regulatory-grade evidence generation. Through joint R&D, participating teams can: • Address domain shift and population variability through cross-national dataset integration; • Establish clinically meaningful performance benchmarks aligned with U.S. healthcare practice; • Generate evidence required for regulatory clearance and reimbursement readiness; • Accelerate global deployment of AI-based diagnostic technologies.
Key Words	1. AI Brain Imaging Analysis 2. Modality Agnostic Model 3. Alzheimer's Pathology Prediction 4. Early Diagnosis 5. Biomarker-based Diagnosis

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (8)
R&D Objectives	Proposals responding to this joint R&D should pursue the following high-level objectives: 1. Development of robust, modality-agnostic AI models for early detection or risk stratification of neurodegenerative diseases; 2. Validation of model performance across multiple institutions, imaging devices, and patient populations; 3. Demonstration of clinical utility, including diagnostic efficiency, interpretability, and workflow compatibility;			

(max 10 pages)

	4. Generation of regulatory-ready evidence supporting future clearance of AI-based diagnostic software. Projects are expected to advance technology maturity from early clinical feasibility (TRL 3–4) toward clinical validation and deployment readiness (TRL 6–8).
R&D Target Specifications	

Target Category	Specific Metric Parameters	Target value (Quantitative)	Verification Method
Data	Training data volume Test data volume	> 10,000 multimodal datasets (Asian cohort) > 1,000 multimodal datasets (U.S. cohort)	Data curation and audit report
Modality-Agnostic Model Performance	Dice index for major brain region	> 0.8 (Inter-modality)	Internal and External Image Analysis
Risk Stratification	Overall AUC	> 0.9 (Aβ positivity) >0.85 (Tau staging) >0.8 (Prognosis)	ROC analysis
	Resolution of intermediate risk (Reduction in ambiguous cases)	≥ 60% reduction achieved by confirmatory testing	Clinical workflow simulation
Clinical usability & validation	Diagnostic Efficiency Improvement	≥ 30% reduction compared to unassisted baseline	Structured reading test
	Regulatory readiness	Complete documentation package for U.S. FDA 510(k) submission	Regulatory Audit & Documentation Review

R&D Plan	While detailed plans should be defined by applicant teams, the anticipated R&D scope may include: • Construction and harmonization of multi-center neuroimaging and clinical datasets; • Development and optimization of AI models capable of handling heterogeneous imaging inputs; • External validation studies in U.S. clinical environments; • Usability, interpretability, and workflow evaluation with clinicians; • Preparatory activities for regulatory submission as a SaMD to the US FDA.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Provide access to U.S. cohort datasets and facilitate external clinical validation to assess robustness across demographic diversity, device variability, and variation in clinical imaging practice in the United States. • Define the scientific and clinical rationale for the intended use, disease stage, model targets, and input features, including selection of imaging signatures and complementary biomarkers aligned with neurodegeneration mechanisms and clinical decision points. • Establish quantifiable, clinically meaningful evaluation metrics and acceptance thresholds, including primary and secondary endpoints, subgroup targets, calibration requirements, and error tolerance to guide model development and interpretation.

	• Support model calibration, domain adaptation, and population specific evaluation to address site and population variability such as ethnicity, comorbidities, acquisition differences, and clinical practice patterns, including cross population benchmarking with age and sex matched comparisons. • Support biomarker and progression modeling using routine clinical MRI, including medial temporal lobe signatures, amyloid related imaging biomarkers when available, and longitudinal trajectory methods to detect early deviation and conversion toward MCI. • Lead scientific dissemination through peer reviewed publications and clinical conference presentations, and when applicable support regulatory readiness and market entry by advising on FDA pathway strategy through predicate device analysis.
Expected R&R of Korean SME Partner	• Lead the development and optimization of the AI-based segmentation model and Alzheimer's risk prediction models using CT and 2D MRI inputs, advancing toward a fully modality-agnostic inference architecture. • Establish and maintain standardized data preprocessing, quality control, and harmonization pipelines to ensure robust and scalable model deployment across diverse imaging protocols. • Conduct initial model training, internal validation, and performance benchmarking using Korean clinical datasets and (or) local partner hospital data. • Apply the Research Institution's established clinical evaluation methodologies and translational research expertise to rigorously assess the clinical efficacy, reproducibility, and generalizability of the Korean SME partner's technology. • Perform domestic pilot studies and prospective evaluations to demonstrate clinical effectiveness and operational feasibility within primary and tertiary care environments in Korea. • Design and execute regulatory-grade clinical validation studies required for U.S. FDA 510(k) submission as a Software as a Medical Device (SaMD), including analytical performance testing and clinical safety and efficacy assessments. • Lead intellectual property (IP) generation and protection activities for newly-developed algorithms, system architectures, and software components.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Outcomes** • Establishment of clinically validated, modality-agnostic AI diagnostic platforms; • Creation of scalable frameworks applicable to multiple neurodegenerative diseases; • Strengthened Korea–US collaborative research ecosystems in digital health. **Commercial and Societal Impact** • Improved access to early diagnostic tools for patients across diverse care settings; • Reduced reliance on invasive and high-cost diagnostic procedures; • Enhanced global competitiveness of Korea–US jointly developed AI medical technologies.

(max 10 pages)

6. References

1. Yan, J.T., et al., *Real-world use of diagnostic tests for mild cognitive impairment, Alzheimer's disease, and other dementias in Medicare fee-for-service beneficiaries.* Alzheimers Dement (Amst), 2025. 17(3): p. e70156.
2. Collij, L.E., et al., *Complementary utility of plasma biomarkers and Abeta-PET for diagnosis, risk-stratification, and treatment monitoring in Alzheimer's disease.* Alzheimers Dement, 2025. 21(10): p. e70763.
3. Yim, S., et al., *CT-derived brain volumes and plasma p-Tau217 for risk stratification of amyloid positivity in early-stage Alzheimer's disease.* Alzheimers Res Ther, 2025. 17(1): p. 233.
4. Lim, K.Y., et al., *Quantifying Brain Atrophy Using a CSF-Focused Segmentation Approach.* Dement Neurocogn Disord, 2025. 24(2): p. 115-125.
5. Lim, K.Y., et al., *Regional CSF volume quantification using deep learning for comparative analysis of brain atrophy in frontotemporal dementia subtypes.* Front Aging Neurosci, 2025. 17: p. 1631640.
6. Chouliaras, L. and J.T. O'Brien, *The use of neuroimaging techniques in the early and differential diagnosis of dementia.* Mol Psychiatry, 2023. 28(10): p. 4084-4097.
7. Therriault, J., et al., *Biomarker modeling of Alzheimer's disease using PET-based Braak staging.* Nat Aging, 2022. 2(6): p. 526-535.
8. Yim, S., et al., *Integrating MRI Volume and Plasma p-Tau217 for Amyloid Risk Stratification in Early-Stage Alzheimer Disease.* Neurology, 2025. 105(6): p. e213954.
9. Park, S., et al., *Computed tomography-based nnU-Net for region-specific brain structural changes across the alzheimer's continuum and frontotemporal dementia subtypes.* Sci Rep, 2025. 15(1): p. 42597.
10. Lteif, D., et al., *Anatomy-Guided, Modality-Agnostic Segmentation of Neuroimaging Abnormalities.* Hum Brain Mapp, 2025. **46**(14): p. e70329.